mse_plots

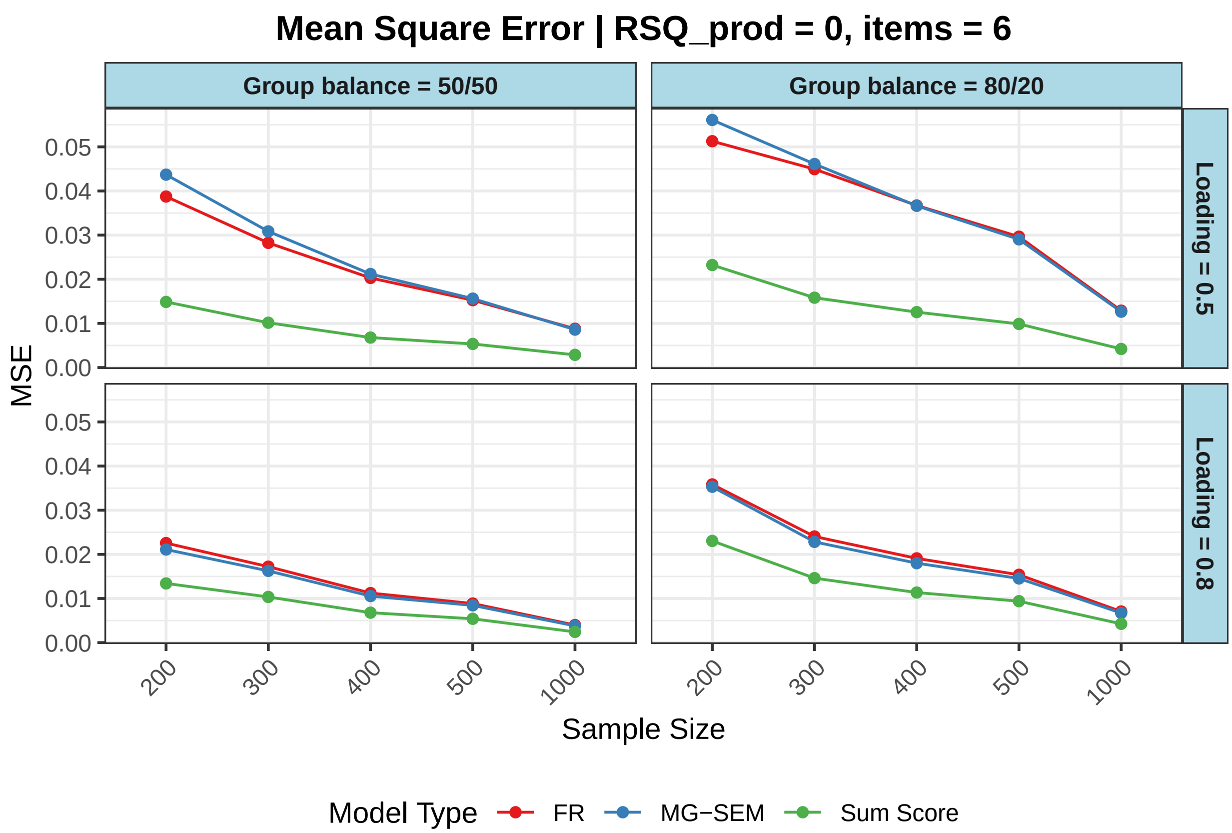
2026-03-30

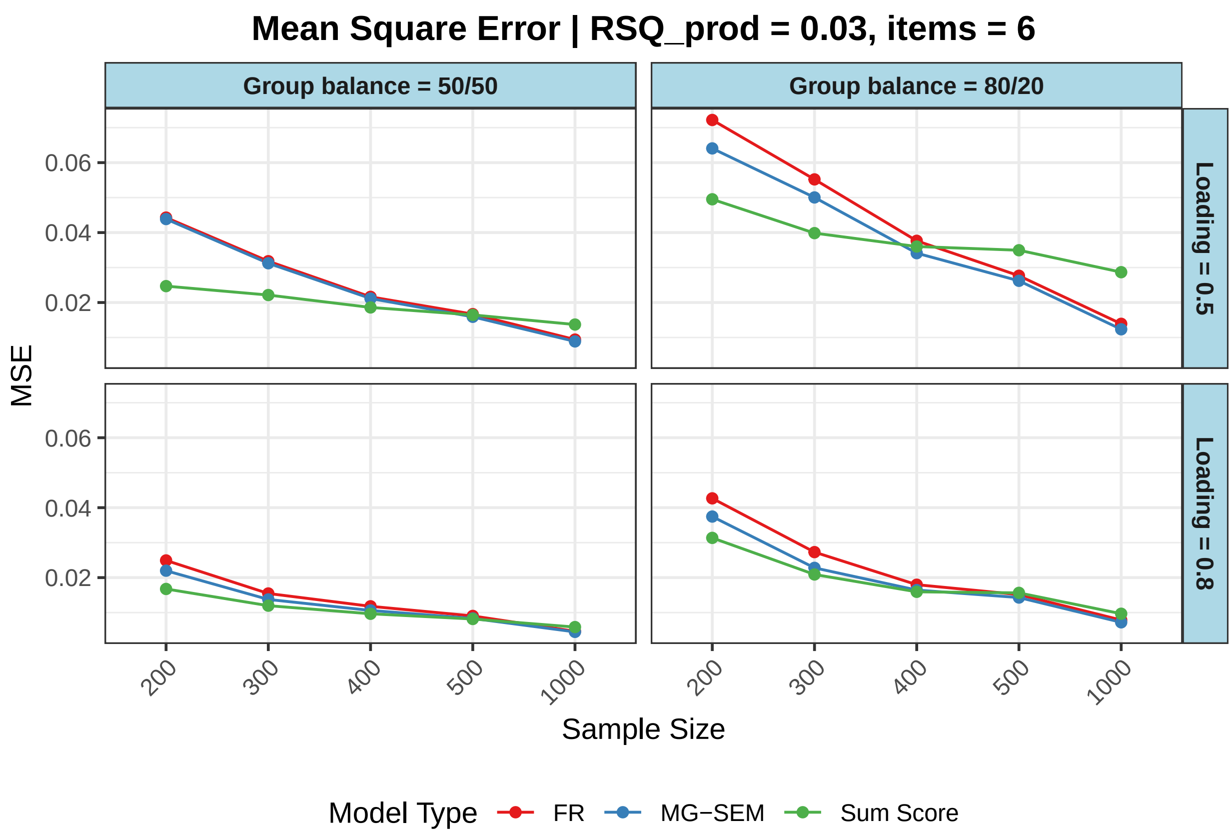
```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

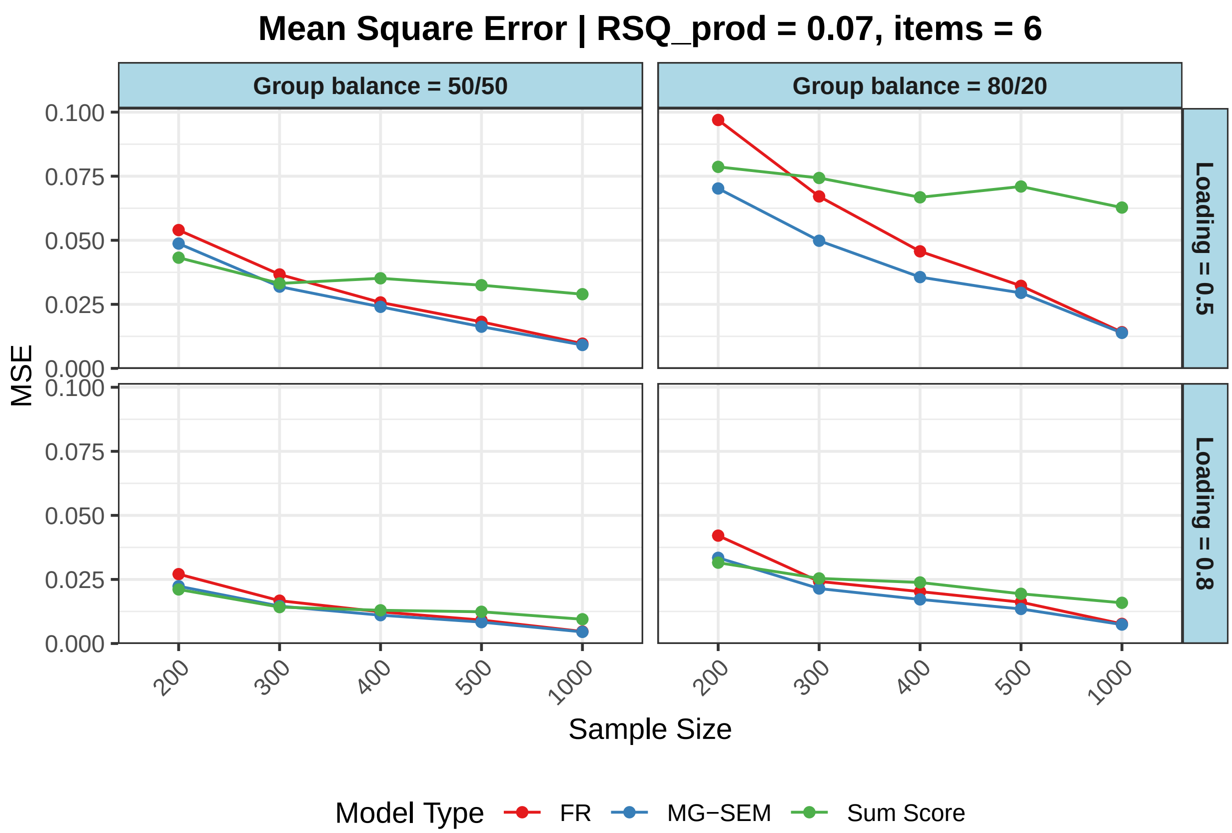
```
options(scipen = 999)


# Load data ----
setwd("/Users/remus/Documents/GitHub/latent_interaction/sim_latent_interaction")
load(file = "agg_results.rda")


# Get unique combinations of rsq_prod and n_items
combinations <- agg_results %>%
  distinct(rsq_prod, n_items)


i=1

# Loop over each combination
for (i in 1:nrow(combinations)) {

  plot_data <- agg_results %>%
    filter(rsq_prod == combinations$rsq_prod[i], n_items  == combinations$n_items[i],
           group_prob %in% c(1,2))

  # Skip if no data
  if (nrow(plot_data) == 0) next

  plot_data <- plot_data %>%
    mutate(
      group_prob = factor(group_prob, levels = c(1, 2)),
      loading    = factor(loading),
```

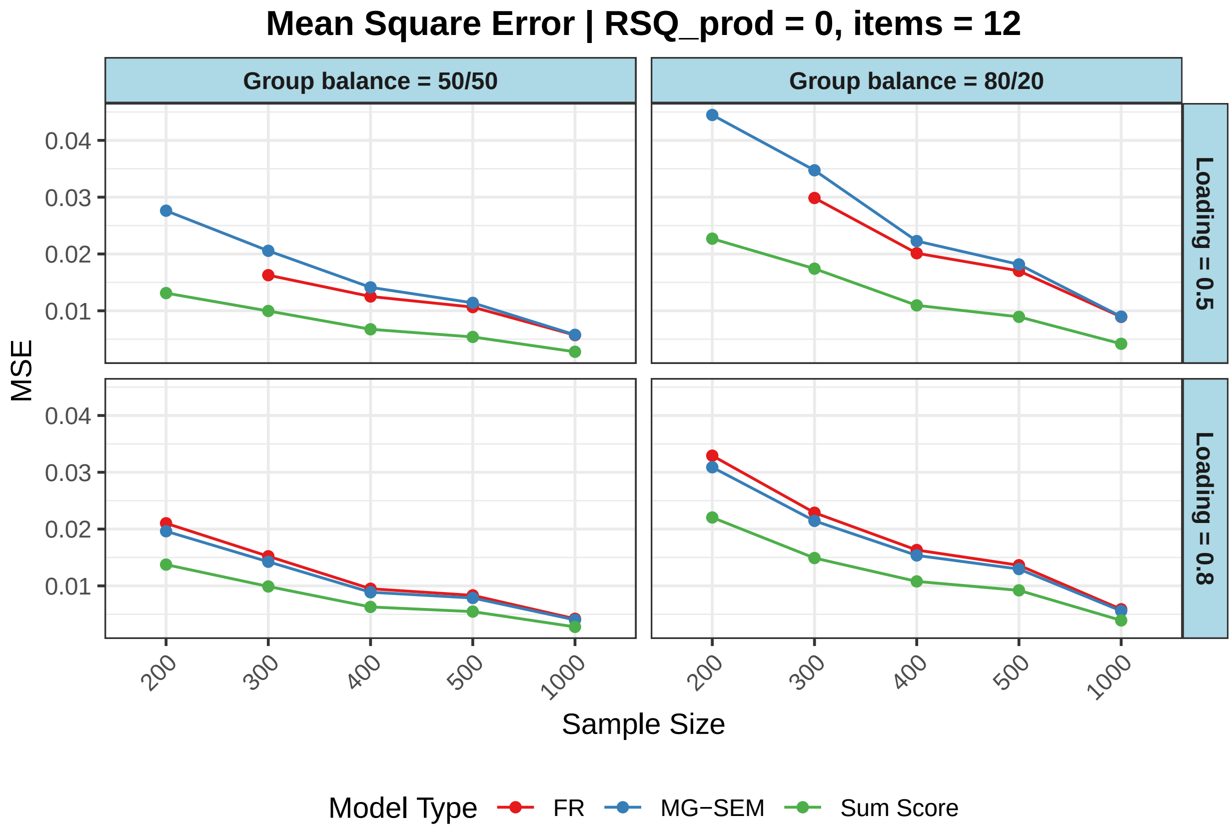
```r
      model_type = factor(model_type, levels = c(1, 2, 3),
                          labels = c("FR", "MG-SEM", "Sum Score")),
      sample_size = factor(sample_size))


  p <- ggplot(plot_data, aes(x = sample_size, y = squared_bias,
                             color = model_type,
                             group = model_type)) +
    geom_line() + geom_point() + facet_grid(rows = vars(loading),
                                            cols = vars(group_prob),
                                            labeller = labeller(
                                              loading = function(x) paste("Loading =", x),
                                              group_prob = function(x) {
                                                x_new <- ifelse(x == 1, "50/50",
                                                                ifelse(x == 2, "80/20", x))
                                                paste("Group balance =", x_new)}))+
    scale_color_brewer(palette = "Set1", name = "Model Type") +
    labs(
      title = paste0("Mean Square Error | RSQ_prod = ", combinations$rsq_prod[i],
                     ", items = ", combinations$n_items[i]),
      x = "Sample Size", y = "MSE") +
    theme_bw() +
    theme(
      plot.title      = element_text(hjust = 0.5, face = "bold"),
      legend.position = "bottom",
      axis.text.x     = element_text(angle = 45, hjust = 1),
      strip.background = element_rect(fill = "lightblue"),
      strip.text       = element_text(face = "bold"))

  print(p)
}
```


Mean Square Error | RSQ_prod = 0, items = 6
Group balance = 50/50
Group balance = 80/20
Loading = 0.5
Loading = 0.8
MSE
0.00
0.01
0.02
0.03
0.04
0.05
200
300
400
500
1000
Sample Size
Model Type
FR
MG−SEM
Sum Score

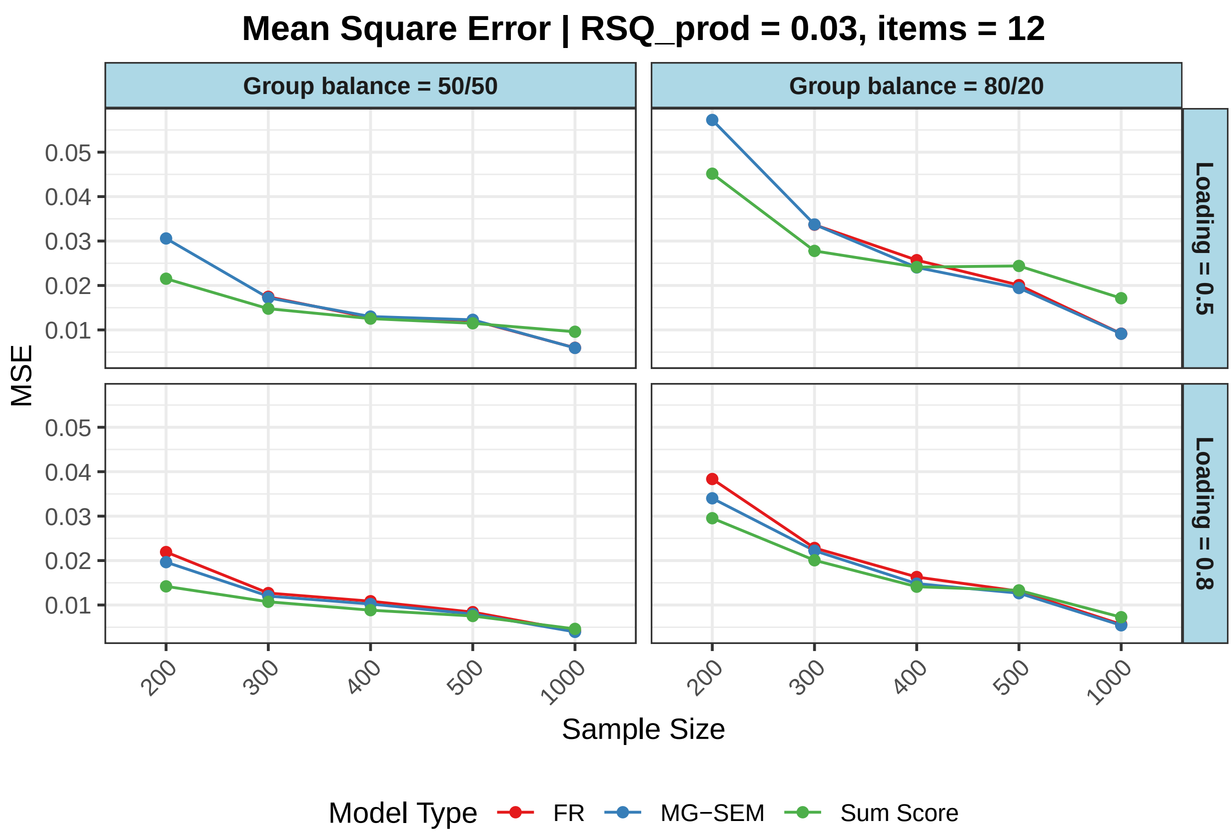


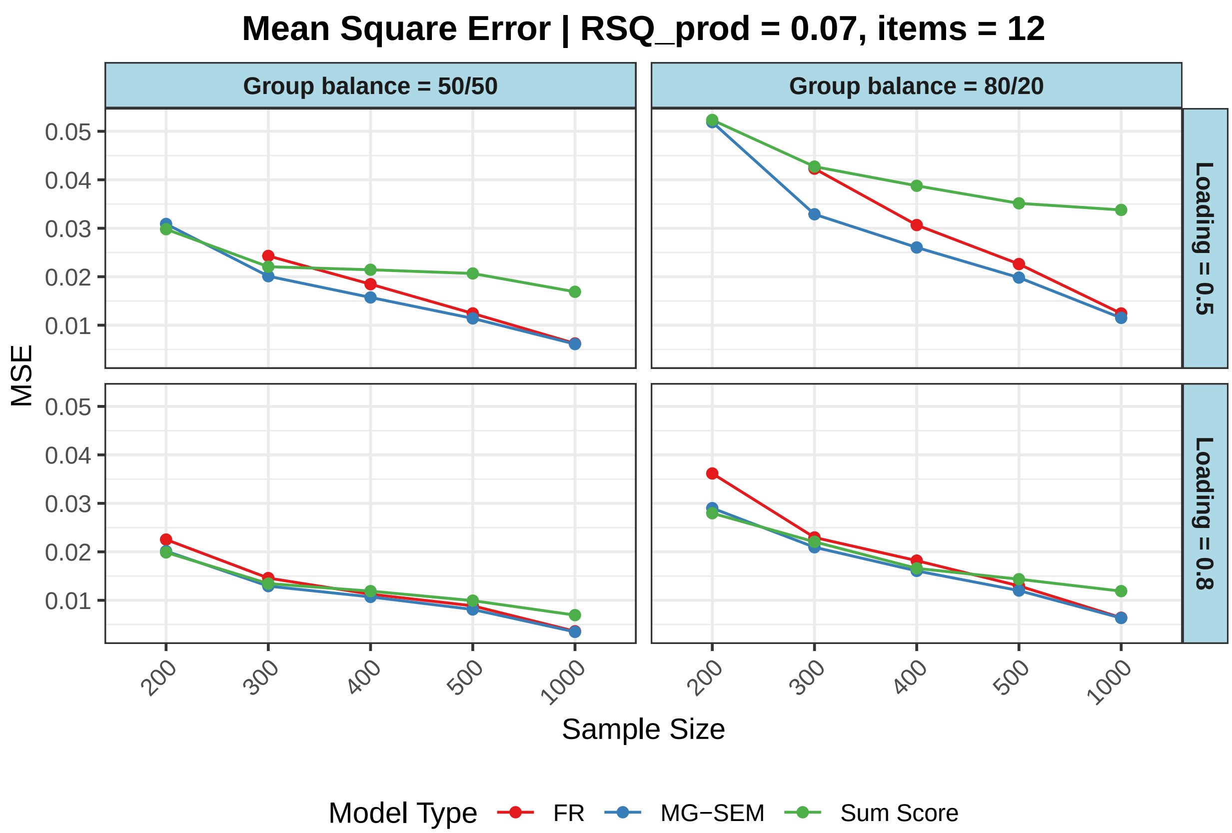
Mean Square Error | RSQ_prod = 0.03, items = 6
Group balance = 50/50
Group balance = 80/20
Loading = 0.5
Loading = 0.8
MSE
0.06
0.04
0.02
200
300
400
500
1000
Sample Size
Model Type
FR
MG−SEM
Sum Score


Mean Square Error | RSQ_prod = 0.07, items = 6
Group balance = 50/50
Group balance = 80/20
Loading = 0.5
Loading = 0.8
MSE
0.000
0.025
0.050
0.075
0.100
200
300
400
500
1000
Sample Size
Model Type
FR
MG−SEM
Sum Score



Mean Square Error | RSQ_prod = 0, items = 12
Group balance = 50/50
Group balance = 80/20
Loading = 0.5
Loading = 0.8
MSE
0.04
0.03
0.02
0.01
200
300
400
500
1000
Sample Size
Model Type
FR
MG−SEM
Sum Score


Mean Square Error | RSQ_prod = 0.03, items = 12
Group balance = 50/50
Group balance = 80/20
Loading = 0.5
Loading = 0.8
MSE
0.05
0.04
0.03
0.02
0.01
200
300
400
500
1000
Sample Size
Model Type
FR
MG−SEM
Sum Score



Mean Square Error | RSQ_prod = 0.07, items = 12
Group balance = 50/50
Group balance = 80/20
Loading = 0.5
Loading = 0.8
MSE
0.05
0.04
0.03
0.02
0.01
200
300
400
500
1000
Sample Size
Model Type
FR
MG−SEM
Sum Score